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MOBILE OBJECT CONTROL DEVICE, MOBILE OBJECT CONTROL METHOD, AND STORAGE MEDIUM

Abstract

A mobile object control device for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway includes a road-type recognition unit configured to recognize whether the mobile object is moving on the roadway or in the prescribed region and a control unit configured to at least partially control a speed of the mobile object. The control unit limits the speed of the mobile object moving on the roadway to a first speed and limits the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.

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Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to a mobile object control device, a mobile object control method, and a storage medium.

BACKGROUND ART

[0002] In the related art, an invention of a single-seater electric vehicle that can move on a sidewalk has been disclosed (Patent Document 1).

CITATION LIST

Patent Document

[0003] [Patent Document 1]

[0004] Japanese Unexamined Patent Application, First Publication No. 2020-189536

SUMMARY OF INVENTION

Technical Problem

[0005] The conventional technology does not assume a mobile object that can move both on the roadway and in a prescribed region different from the roadway. For this reason, it may be difficult to perform appropriate control both on the roadway and in the prescribed region different from the roadway.

[0006] The present invention has been made in consideration of such circumstances, and an objective of the present invention is to provide a mobile object control device, a mobile object control method, and a storage medium capable of performing appropriate control both on a roadway and in a prescribed region different from the roadway.

Solution to Problem

[0007] A mobile object control device, a mobile object control method, and a storage medium according to the present invention adopt the following configurations.

[0008] (1): According to an aspect of the present invention, there is provided a mobile object control device for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway, the mobile object control device including: a road-type recognition unit configured to recognize whether the mobile object is moving on the roadway or in the prescribed region; and a control unit configured to at least partially control a speed of the mobile object, the control unit limiting the speed of the mobile object moving on the roadway to a first speed and limiting the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.

[0009] (2): In the above-described aspect (1), the road-type recognition unit causes an external notification device to provide a notification indicating that the mobile object is moving in the prescribed region to the outside of the mobile object when it is recognized that the mobile object is moving in the prescribed region.

[0010] (3): In the above-described aspect (1) or (2), the road-type recognition unit recognizes whether the mobile object is moving on the roadway or moving in the prescribed region on the basis of an output of an external world detection device for detecting a situation outside of the mobile object.

[0011] (4): In any one of the above-described aspects (1) to (3), the road-type recognition unit recognizes whether the mobile object is moving on the roadway or moving in the prescribed region on the basis of an operation of an occupant for a switch provided on the mobile object.

[0012] (5): In any one of the above-described aspects (1) to (4), the control unit receives a speed instruction from an occupant via an instruction reception unit provided on the mobile object, automatically controls at least the speed of the mobile object moving on the roadway with reference to the speed instruction, and limits the speed of the mobile object moving on the roadway to the first speed regardless of the speed instruction that has been designated.

[0013] (6): In any one of the above-described aspects (1) to (5), the control unit receives a speed instruction from an occupant via an instruction reception unit provided on the mobile object, automatically controls at least the speed of the mobile object moving in the prescribed region with reference to the speed instruction, and limits the speed of the mobile object moving in the prescribed region to the second speed regardless of the speed instruction that has been designated.

[0014] (7): In the above-described aspect (5) or (6), the instruction reception unit is a dial switch for issuing an instruction to increase or decrease the speed according to a rotation angle.

[0015] (8): In any one of the above-described aspects (1) to (7), a first operation element for receiving an acceleration or deceleration instruction and a second operation element for receiving a steering instruction are provided on the mobile object, and the control unit switches a mode to one of mode A in which acceleration, deceleration, and steering of the mobile object are controlled on the basis of an operation on one of the first operation element and the second operation element, mode B in which the acceleration, deceleration, and steering of the mobile object are controlled on the basis of an operation on both the first operation element and the second operation element, and mode C in which the acceleration, deceleration, and steering of the mobile object are controlled without being based on an operation on either the first operation element or the second operation element.

[0016] (9): According to another aspect of the present invention, there is provided a mobile object control method including: recognizing, by a computer for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway, whether the mobile object is moving on the roadway or in the prescribed region; at least partially controlling, by the computer, a speed of the mobile object; and limiting, by the computer, the speed of the mobile object moving on the roadway to a first speed and limiting the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.

[0017] (10): According to yet another aspect of the present invention, there is provided a storage medium storing a program for causing a computer for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway to: recognize whether the mobile object is moving on the roadway or in the prescribed region; at least partially control a speed of the mobile object; and limit the speed of the mobile object moving on the roadway to a first speed and limit the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed. The storage medium may be a computer-readable non-transitory storage medium.

Advantageous Effects of Invention

[0018] According to aspects (1) to (10), it is possible to perform appropriate control both on a roadway and in a prescribed region different from the roadway.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0019] FIG. 1 is a diagram showing an example of a configuration of a mobile object and a control device according to an embodiment.

[0020] FIG. 2 is a diagram showing an example of an external configuration of a dial switch.

[0021] FIG. 3 is a perspective view of the mobile object seen from above.

[0022] FIG. 4 is a diagram showing an example of a situation where specific speed control is performed.

[0023] FIG. 5 is a flowchart showing an example of a flow of a process executed by the control device of the embodiment.

DESCRIPTION OF EMBODIMENTS

[0024] Hereinafter, a mobile object control device, a mobile object control method, and a storage medium according to the present invention will be described with reference to the drawings. A mobile object moves both on a roadway and in a prescribed region different from the roadway. The mobile object may be a vehicle that an occupant can board or may be an autonomous mobile object capable of performing an unmanned autonomous travel process. The latter autonomous mobile object is used, for example, for transporting luggage and the like. In the following description, it is assumed that one or more occupants are in the mobile object. The prescribed region is, for example, a sidewalk. Also, the prescribed region may be a part or all of a roadside strip, a bicycle lane, a public open space, or the like or may include all sidewalks, roadside strips, bicycle lanes, public open spaces, and the like. In the following description, the prescribed region is assumed to be a sidewalk. The part described as the “sidewalk” in the following description can be read as a “prescribed region” as appropriate.

First Embodiment

[0025] FIG. 1 is a diagram showing an example of a configuration of a mobile object **1** and a control device **100** according to the embodiment. The mobile object **1** includes, for example, an external world detection device **10**, a mobile object sensor **12**, an operation element **14**, an internal camera **16**, a positioning device **18**, a dialogue device **20**, a mode changeover switch **22**, a dial switch **24**, a movement mechanism **30**, a drive device **40**, an external notification device **50**, a storage device **70**, and a control device **100**. Among these configurations, some configurations that are not essential for implementing the functions of the present invention may be omitted.

[0026] The external world detection device **10** is one of various types of devices whose detection ranges are in a travel direction of the mobile object **1**. The external world detection device **10** includes an external camera, a radar device, a light detection and ranging (LIDAR) sensor, a sensor fusion device, and the like. The external world detection device **10** outputs information (an image, a position of a physical object, or the like) indicating a detection result to the control device **100**.

[0027] The mobile object sensor **12** includes, for example, a speed sensor, an acceleration sensor, a yaw rate (angular speed) sensor, a direction sensor, an operation amount detection sensor attached to the operation element **14**, and the like. The operation element **14** includes, for example, an operation element for issuing an acceleration/deceleration instruction (for example, an accelerator pedal or a brake pedal) and an operation element for issuing a steering instruction (for example, a steering wheel). In this case, the mobile object sensor **12** may include an accelerator opening degree sensor, a brake pedal depression amount sensor, a steering torque sensor, and the like. The mobile object **1** may include an operation element of an aspect other than the above (for example, a non-annular rotation operation element, a joystick, a button, or the like) as the operation element **14**.

[0028] The internal camera **16** images at least the head of the occupant of the mobile object **1** from the front. The internal camera **16** is a digital camera that uses an imaging element such as a charge-coupled device (CCD) or a complementary metal oxide semiconductor (CMOS). The internal camera **16** outputs the captured image to the control device **100**.

[0029] The positioning device **18** is a device for measuring a position of the mobile object **1**. The positioning device **18** is, for example, a global navigation satellite system (GNSS) receiver and identifies the position of the mobile object **1** on the basis of a signal received from a GNSS satellite and outputs the identified position as position information. Also, the position information of the

mobile object **1** may be estimated from a position of a Wi-Fi base station to which the communication device to be described below is connected.

[0030] The dialogue device **20** includes, for example, a speaker, a microphone, a touch panel, a communication device, and the like. The dialogue device **20** appropriately processes a sound of the occupant collected by the microphone, transmits the processed sound to a server device via a network using a communication device, and provides information of the sound from a speaker on the basis of the information returned from the server device. The dialogue device **20** may be referred to as an agent device, a concierge device, an assistance device, or the like. The server device has a sound recognition function, a natural language processing function, a semantic interpretation function, a reply content decision function, and the like. Also, the dialogue device **20** transmits the position information to the server device and the server device may return the position information and corresponding information of a facility in response to a guidance request issued by the occupant (for example, “Is there a good ramen shop around here?”). In this case, voice guidance such as “There is one at the point where you turn left ahead” is provided by the dialogue device **20**. In addition to this, the dialogue device **20** has a function of receiving natural utterances of the occupant and returning an appropriate reply. Also, the dialogue device **20** has a function of performing simple dialogue without going through the server device, such as asking a question from the device side and receiving a reply, and may ask the occupant a question in response to a request from the control device **100**. The dialogue device **20** is an example of an interface device.

[0031] The mode changeover switch **22** is a switch to be operated by an occupant. The mode changeover switch **22** may be a mechanical switch or a graphical user interface (GUI) switch set on a touch panel. The mode changeover switch **22**, for example, receives an operation for switching a driving mode to any one of mode A, which is an assist mode in which one of a steering operation and acceleration/deceleration control is performed by the occupant and the other is automatically performed and which may be mode A-1 in which the steering operation is performed by the occupant and the acceleration/deceleration control is performed automatically and mode A-2 in which an acceleration/deceleration operation is performed by the occupant and steering control is performed automatically, mode B, which is a manual driving mode in which the steering operation and the acceleration/deceleration operation are performed by the occupant, and mode C, which is an automated driving mode in which the steering control and the acceleration/deceleration control are performed automatically.

[0032] FIG. **2** is a diagram showing an example of an external configuration of the dial switch **24**. The dial switch **24** is installed at any position where an occupant P in the mobile object **1** can perform an operation and can be rotated clockwise and counterclockwise. For example, a clockwise direction is set to indicate acceleration and a counterclockwise direction is set to indicate deceleration. The dial switch **24** can change the speed by, for example, 5 [km/h] every time a scale is operated. Details of speed control using the dial switch **24** will be described below.

[0033] The movement mechanism **30** is a mechanism for moving the mobile object **1** on a road. The movement mechanism **30** is, for example, a wheel group including a steered wheel and a drive wheel. Also, the movement mechanism **30** may be a leg for multi-legged walking.

[0034] The drive device **40** outputs a force to the movement mechanism **30** and moves the mobile object **1**. For example, the drive device **40** includes a motor for driving the drive wheel, a battery for storing electric power supplied to the motor, a steering device for adjusting a steering angle of the steered wheel, and the like. The drive device **40** may include an internal combustion engine, a fuel cell, or the like as a driving force output means or a power generation means. Also, the drive device **40** may further include a brake device due to friction force or air resistance.

[0035] The external notification device **50** is, for example, a lamp, a display device, a speaker, or the like provided on an outer plate part of the mobile object **1** and configured to provide an information notification to the outside of the mobile object **1**. The external notification device **50** performs different operations in a state in which the mobile object **1** is moving on the sidewalk and

a state in which the mobile object **1** is moving on the roadway. For example, the external notification device **50** is controlled so that light of a lamp is emitted when the mobile object **1** is moving on the sidewalk and the light of the lamp is not emitted when the mobile object **1** is moving on the roadway. A color of light emission of this lamp is preferably a color defined by law. The external notification device **50** may be controlled so that the lamp emits light in green when the mobile object **1** is moving on the sidewalk and the lamp emits light in blue when the mobile object **1** is moving on the roadway. When the external notification device **50** is a display device, the external notification device **50** displays text or graphics indicating that “the mobile object **1** is traveling on the sidewalk” when the mobile object **1** is traveling on the sidewalk.

[0036] FIG. **3** is a perspective view of the mobile object **1** seen from above. In FIG. **3**, FW denotes a steered wheel, RW denotes a drive wheel, SD denotes a steering device, MT denotes a motor, and BT denotes a battery. The steering device SD, the motor MT, and the battery BT are included in the drive device **40**. Also, AP denotes an accelerator pedal, BP denotes a brake pedal, WH denotes a steering wheel, SP denotes a speaker, and MC denotes a microphone. The mobile object **1** shown in FIG. **3** is a single-seater mobile object and the occupant P sits in a driver's seat DS and wears a seat belt SB. An arrow D1 is a travel direction (speed vector) of the mobile object **1**. The external world detection device **10** is provided near a front end of the mobile object **1**, the internal camera **16** is at a position where the head of the occupant P can be imaged from the front of the occupant P, and the mode changeover switch **22** is provided at a boss portion of the steering wheel WH. Also, the external notification device **50** as a display device is provided near the front end of the mobile object **1**.

[0037] Returning to FIG. **1**, the storage device **70** is a non-transitory storage device such as, for example, a hard disk drive (HDD), a flash memory, or a random-access memory (RAM). The storage device **70** stores map information **72**, a program **74** to be executed by the control device **100**, and the like. Although the storage device **70** is shown outside of a frame of the control device **100** in FIG. **1**, the storage device **70** may be included in the control device **100**.

[Control Device]

[0038] The control device **100** includes, for example, a road-type recognition unit **120**, a physical object recognition unit **130**, and a control unit **140**. For example, these components are implemented by a hardware processor such as a central processing unit (CPU) executing the program (software) **74**. Also, some or all of these components may be implemented by hardware (including a circuit; circuitry) such as a large-scale integration (LSI) circuit, an application-specific integrated circuit (ASIC), a field-programmable gate array (FPGA), or a graphics processing unit (GPU) or may be implemented by software and hardware in cooperation. The program may be pre-stored in the storage device **70** or may be stored in a removable storage medium (a non-transitory storage medium) such as a DVD or a CD-ROM and installed in the storage device **70** when the storage medium is mounted in a drive device.

[0039] The road-type recognition unit **120** recognizes whether the mobile object **1** is moving on the roadway or the sidewalk. For example, the road-type recognition unit **120** recognizes whether the mobile object **1** is moving on the roadway or the sidewalk by analyzing an image captured by an external camera of the external world detection device **10**. An example of image analysis is semantic segmentation. The road-type recognition unit **120** classifies each pixel in the frame of the image as a class (a roadway, a sidewalk, a boundary, a physical object, or the like) to assign a label, recognizes that the mobile object **1** is moving on the roadway when the number of pixels to which the label of the roadway is assigned is larger in a region corresponding to the front of the mobile object **1**, and recognizes that the mobile object **1** is moving on the sidewalk when the number of pixels to which the label of the sidewalk is assigned is larger in the region corresponding to the front of the mobile object **1** in the image. In addition to this, the road-type recognition unit **120** may recognize that the mobile object **1** is moving on the roadway when a vehicle is recognized in the region corresponding to the front of the mobile object **1** in the image and recognize that the mobile

object **1** is moving on the sidewalk when a pedestrian is recognized in the region corresponding to the front of the mobile object **1** in the image. Also, the road-type recognition unit **120** may recognize that the mobile object **1** is moving on the roadway when a width of a road surface region in the region corresponding to the front of the mobile object **1** in the image is larger and recognize that the mobile object **1** is moving on the sidewalk when the width of the road surface region in the region corresponding to the front of the mobile object **1** in the image is smaller. Also, the road-type recognition unit **120** may compare the position information of the mobile object **1** with the map information **72** and recognize whether the mobile object **1** is moving on the roadway or the sidewalk. In this case, the map information needs to have the accuracy of a degree to which the sidewalk and the roadway can be distinguished from position coordinates. When the “prescribed region” is not only a sidewalk, the road-type recognition unit **120** also performs a similar process for roadside strips, bicycle lanes, open spaces, and the like.

[0040] The physical object recognition unit **130** recognizes a physical object located near the mobile object **1** on the basis of an output of the external world detection device **10**. Physical objects include some or all of mobile objects such as a vehicle, a bicycle, and a pedestrian, runway boundaries such as a road marking, a step, a guardrail, a shoulder, and a median strip, structures installed on the road such as a road sign and a signboard, and an obstacle such as a falling object that exists (falls) on the runway. For example, an acquisition unit **110** acquires information of the presence, position, and type of another mobile object by inputting an image captured by the external camera to a trained model trained to output the information of the presence, position, and type of a physical object when the image captured by the external camera of the external world detection device **10** is input. Other types of mobile objects can also be estimated on the basis of a size in the image, an intensity of a reflected wave received by a radar device of the external world detection device **10**, and the like. Also, the acquisition unit **110** acquires the speed of another mobile object detected by the radar device using, for example, a Doppler shift or the like.

[0041] The control unit **140** controls, for example, the drive device **40** in accordance with a set operation mode.

[0042] In mode A-1, the control unit **140** refers to information of the runway and the physical object based on the output of the physical object recognition unit **130** and controls the motor MT of the drive device **40** so that a distance from the physical object located in front of the mobile object **1** is maintained at a certain distance or longer when the mobile object **1** moves on the roadway and the mobile object **1** moves at a first speed V1 (for example, a speed greater than or equal to 10 [km/h] and less than or equal to several tens of kilometers per hour [km/h]) when the distance from the physical object located in front of the mobile object **1** is sufficiently long. The control unit **140** controls the motor MT of the drive device **40** so that a certain distance to the physical object located in front of the mobile object **1** is maintained at the certain distance or longer when the mobile object **1** moves on the sidewalk and the mobile object **1** moves at a second speed V2 (for example, a speed lower than 10 [km/h]) when the distance from the physical object located in front of the mobile object **1** is sufficiently long. This function is similar to an adaptive cruise control (ACC) function of a vehicle with the first speed V1 or the second speed V2 set as a set speed and the technology used in ACC can be used. Also, in mode A-1, the control unit **140** controls the steering device SD so that a steering angle of the steered wheel is changed on the basis of an operation amount of the operation element **14** such as the steering wheel. This function is similar to that of the power steering device and the technology used in the power steering device can be used. Also, electronic control is not performed in relation to steering and the mobile object **1** may include a steering device in which the operation element **14** and the steering mechanism are mechanically connected.

[0043] In mode A-2, the control unit **140** refers to information of the runway and the physical object based on the output of the physical object recognition unit **130**, generates a target trajectory along which the mobile object **1** can move by avoiding the physical object within the runway, and

controls the steering device SD of the drive device **40** so that the mobile object **1** moves along the target trajectory. In relation to acceleration/deceleration, the control unit **140** controls the motor MT of the drive device **40** on the basis of the speed of the mobile object **1** and the operation amount of the accelerator pedal or the brake pedal. The control unit **140** controls the motor MT of the drive device **40** using the first speed V1 as an upper limit speed (in the case of mode A-2, when the upper limit speed is reached, it indicates that the mobile object **1** is not accelerated even if there is a further acceleration instruction) when the mobile object **1** is moving on the roadway and controls the drive device **40** using the second speed V2 as the upper limit speed when the mobile object **1** is moving on the sidewalk.

[0044] In mode B, the control unit **140** controls the motor MT of the drive device **40** on the basis of the speed of the mobile object **1** and the operation amount of the accelerator pedal or the brake pedal. The control unit **140** controls the motor MT of the drive device **40** using the first speed V1 as the upper limit speed (in the case of mode B, when the upper limit speed is reached, it indicates that the mobile object **1** is not accelerated even if there is a further acceleration instruction) when the mobile object **1** is moving on the roadway and controls the motor MT of the drive device **40** using the second speed V2 as the upper limit speed when the mobile object **1** is moving on the sidewalk. In relation to steering, it is similar to mode A-1.

[0045] In mode C, the control unit **140** refers to information of a runway and a physical object based on the output of the physical object recognition unit **130**, generates a target trajectory along which the mobile object **1** can move by avoiding the physical object within the runway, and controls the drive device **40** so that the mobile object **1** moves along the target trajectory. In mode C as well, the control unit **140** controls the drive device **40** using the first speed V1 as the upper limit speed when the mobile object **1** is moving on the roadway and controls the drive device **40** using the second speed V2 as the upper limit speed when the mobile object **1** is moving on the sidewalk.

[Control According to Operation of Dial Switch]

[0046] The control unit **140** controls the speed of the mobile object **1** on the basis of an operation amount for the dial switch **24** in modes (modes A-1 and mode C) in which the speed is controlled regardless of the operation amount of the accelerator pedal or the brake pedal. When the mobile object **1** moves on the roadway in modes A-1 and C, the control unit **140** sets a target speed according to a situation. For example, the control unit **140** corrects a default target speed Vdef on the basis of an average speed of a road, the number of traffic participants, a distance to a physical object around the mobile object **1**, and the like, and decides on a target speed Vref according to the situation. The default target speed Vdef is preset, for example, to a speed lower than the first speed V1. Further, the control unit **140** decides on a final target speed V # by correcting the target speed Vref according to the situation on the basis of an operation position of the dial switch **24**. Here, when a value obtained by correcting the target speed Vref according to the situation on the basis of the operation position of the dial switch **24** exceeds the first speed V1, the control unit **140** sets the final target speed V # as the first speed V1. Thus, the control unit **140** receives a speed instruction from the occupant via an instruction reception unit such as the dial switch **24** provided on the mobile object **1** (something having a property of receiving an additional instruction for a current speed without receiving an instruction connected to the output torque such as the accelerator pedal). Also, the control unit **140** automatically controls the speed at least when the mobile object **1** moves on the roadway with reference to the additional instruction. However, the control unit **140** limits the speed at which the mobile object **1** moves on the roadway to the first speed V1 regardless of the speed instruction that has been designated. Thereby, for example, it is possible to comply with laws and regulations of a country where the mobile object **1** moves.

[0047] FIG. 4 is a diagram showing a change in control content based on an operation amount for the dial switch **24**. In FIG. 4, it is assumed that the default target speed Vdef=20 [km/h] and the first speed V1=30 [km/h]. In FIG. 4, in the situation shown in (1), because there is a vehicle on the

travel direction side of the mobile object **1** and the control unit **140** determines to suppress acceleration according to its presence, the target speed V_{ref} according to the situation is decided on to be the same as the default target speed V_{def} . Because the operation position of the dial switch **24** is zero, the final target speed $V \#$ is also the same as the default target speed V_{def} .

[0048] In the situation shown in the following (2), because the road situation does not change from the situation shown in (1), the target speed V_{ref} according to the situation is decided on to be the same as the default target speed V_{def} , but the operation position of the dial switch **24** is changed to +5, such that the final target speed $V \#$ has a value greater than the default target speed V_{def} by 5 [km/h].

[0049] In the situation shown in the following (3), because the vehicle in front of the mobile object **1** is no longer recognized by the mobile object **1** due to a lane change or the like, the control unit **140** raises the target speed V_{ref} according to the situation to the first speed $V1$. Although the operation position of the dial switch **24** is maintained at +5, the corrected speed exceeds the first speed $V1$ when the target speed V_{ref} according to the situation is corrected by +5, such that the final target speed $V \#$ is limited to the first speed $V1$.

[0050] In the situation shown in the following (4), the front of the mobile object **1** is congested and the control unit **140** sets the target speed V_{ref} according to the situation to a value less than the default target speed V_{def} . On the other hand, the operation position of the dial switch **24** is +5 and a value obtained by correcting the target speed V_{ref} according to the situation by +5 is set as the final target speed $V \#$.

[0051] FIG. 5 is a flowchart showing an example of a flow of a process executed by the control device **100** of the embodiment. In the process of the present flowchart, mode A is assumed. The process of the present flowchart is iteratively executed, for example, at prescribed intervals.

[0052] First, the road-type recognition unit **120** determines whether or not the mobile object **1** is moving on the roadway (whether the mobile object **1** is moving on the sidewalk) (step S200).

When it is determined that the mobile object **1** is moving on the roadway, the control unit **140** decides on the target speed V_{ref} according to the situation by correcting the default target speed V_{def} on the basis of an average speed of the road, the number of traffic participants, a distance from a physical object near the mobile object **1**, and the like as described above (step S202) and decides on the final target speed $V \#$ by correcting the default target speed V_{def} according to the situation on the basis of the operation position of the dial switch **24** (step S204). Subsequently, the control unit **140** determines whether or not the final target speed $V \#$ exceeds the first speed $V1$ (step S206). When the final target speed $V \#$ exceeds the first speed $V1$, the control unit **140** limits the final target speed $V \#$ to the first speed $V1$ (step S208). Also, the control unit **140** moves the mobile object **1** at the final target speed $V \#$ (step S210).

[0053] Also, when it is determined in step S200 that the mobile object **1** is moving on the sidewalk, the control unit **140** performs control of a case where the mobile object **1** is moving on the sidewalk (step S212).

[0054] The above-described “control according to the operation of the dial switch” may be performed not only when the mobile object **1** is moving on the roadway but also when the mobile object **1** is moving on the sidewalk. In this case, the control unit **140** decides on the target speed V_{ref} according to the situation and decides on the final target speed $V \#$ by correcting the target speed V_{ref} according to the situation on the basis of the operation position of the dial switch **24** when the mobile object **1** is moving on the sidewalk and sets the final target speed $V \#$ as the second speed $V2$ when a value obtained by correcting the target speed V_{ref} according to the situation on the basis of the operation position of the dial switch **24** exceeds the second speed $V2$. Although it is assumed that control according to the operation of the dial switch is executed only for the roadway in the flowchart of FIG. 5, processing similar to steps S202 to S210 (in this regard, $V1$ is replaced with $V2$) is performed even if the mobile object **1** is moving on the sidewalk in a case where control according to an operation of the dial switch is performed even if the mobile

object **1** is moving on the sidewalk.

[0055] According to the first embodiment described above, there are provided the road-type recognition unit **120** configured to recognize whether the mobile object **1** is moving on a roadway or a sidewalk and the control unit **140** configured to at least partially control a speed of the mobile object **1**, wherein the control unit **140** limits the speed of the mobile object **1** moving on the roadway to the first speed **V1** and limits the speed of the mobile object **1** moving on the sidewalk to the second speed **V2** that is lower than the first speed **V1**, whereby it is possible to perform appropriate control on both the roadway and the sidewalk.

Second Embodiment

[0056] Hereinafter, a second embodiment will be described. The control device **100** of the second embodiment is different from that of the first embodiment in the function of the road-type recognition unit **120**. The road-type recognition unit **120** of the second embodiment, for example, recognizes whether the mobile object **1** is traveling on the roadway or on the sidewalk in accordance with the operation of the occupant performed on a road-type input switch (not shown) provided on the mobile object. The road-type input switch is provided, for example, in the boss portion of the steering wheel **WH** or the like. The road-type input switch is, for example, operable up and down, has a mechanism for holding a position where an operation is performed, and indicates a roadway when it is operated on the upper side, and indicates a sidewalk when it is operated on the lower side. Also, the road-type input switch is provided in the form of a button or a GUI switch, may be switched for recognition during traveling on the sidewalk when an operation is performed in a state in which the control device **100** is recognizing traveling on the roadway, and may be switched for recognition during traveling on the roadway when an operation is performed in a state in which the control device **100** is recognizing traveling on the sidewalk. According to the configuration, it is possible to omit a function of automatically recognizing whether the mobile object **1** is traveling on the roadway or the sidewalk and reduce the processing load and the cost. In this regard, because there is a possibility of an erroneous operation of the occupant, the external notification device **50** preferably externally provides an information notification in the second embodiment.

<Others>

[0057] In each of the above embodiments, the control unit **140** may execute only control of either mode A or mode B. That is, the mobile object **1** may execute the assist mode without executing the manual mode or may execute the manual mode without executing the assist mode. Also, the control unit **140** executes control of both mode A and mode B, but may perform specific speed control only when either mode A or mode B is executed. In either case, it may be arbitrarily determined whether to perform the control of mode C.

[0058] Also, the mobile object **1** may not have an accelerator pedal and may receive speed instructions exclusively via the dial switch **24**. In this case, mode A-2 and mode B are not executed and only mode A-1 and mode C are executed substantially.

[0059] The embodiment described above can be represented as follows.

[0060] A mobile object control device including: [0061] a hardware processor; and [0062] a storage device storing a program, [0063] wherein the hardware processor executes the program stored in the storage device to: [0064] recognize whether a mobile object is moving on a roadway or in a prescribed region different from the roadway; [0065] at least partially control a speed of the mobile object; and [0066] limit the speed of the mobile object moving on the roadway to a first speed and limit the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.

[0067] Although modes for carrying out the present invention have been described above using embodiments, the present invention is not limited to the embodiments and various modifications and substitutions can also be made without departing from the scope and spirit of the present invention.

REFERENCE SIGNS LIST

[0068] **10** External world detection device [0069] **12** Mobile object sensor [0070] **14** Operation element [0071] **16** Internal camera [0072] **18** Positioning device [0073] **20** Dialogue device [0074] **22** Mode changeover switch [0075] **24** Dial Switch [0076] **30** Movement mechanism [0077] **40** Drive device [0078] **50** External notification device [0079] **70** Storage device [0080] **100** Control device [0081] **120** Road-type recognition unit [0082] **130** Physical object recognition unit [0083] **140** Control unit

Claims

1. A mobile object control device for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway, the mobile object control device comprising: a road-type recognition unit configured to recognize whether the mobile object is moving on the roadway or in the prescribed region; and a control unit configured to at least partially control a speed of the mobile object, the control unit limiting the speed of the mobile object moving on the roadway to a first speed and limiting the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.
2. The mobile object control device according to claim 1, wherein the road-type recognition unit causes an external notification device to provide a notification indicating that the mobile object is moving in the prescribed region to the outside of the mobile object when it is recognized that the mobile object is moving in the prescribed region.
3. The mobile object control device according to claim 1, wherein the road-type recognition unit recognizes whether the mobile object is moving on the roadway or moving in the prescribed region on the basis of an output of an external world detection device for detecting a situation outside of the mobile object.
4. The mobile object control device according to claim 1, wherein the road-type recognition unit recognizes whether the mobile object is moving on the roadway or moving in the prescribed region on the basis of an operation of an occupant for a switch provided on the mobile object.
5. The mobile object control device according to claim 1, wherein the control unit receives a speed instruction from an occupant via an instruction reception unit provided on the mobile object, automatically controls at least the speed of the mobile object moving on the roadway with reference to the speed instruction, and limits the speed of the mobile object moving on the roadway to the first speed regardless of the speed instruction that has been designated.
6. The mobile object control device according to claim 1, wherein the control unit receives a speed instruction from an occupant via an instruction reception unit provided on the mobile object, automatically controls at least the speed of the mobile object moving in the prescribed region with reference to the speed instruction, and limits the speed of the mobile object moving in the prescribed region to the second speed regardless of the speed instruction that has been designated.
7. The mobile object control device according to claim 5, wherein the instruction reception unit is a dial switch for issuing an instruction to increase or decrease the speed according to a rotation angle.
8. The mobile object control device according to claim 1, wherein a first operation element for receiving an acceleration or deceleration instruction and a second operation element for receiving a steering instruction are provided on the mobile object, and wherein the control unit switches a mode to one of mode A in which acceleration, deceleration, and steering of the mobile object are controlled on the basis of an operation on one of the first operation element and the second operation element, mode B in which the acceleration, deceleration, and steering of the mobile object are controlled on the basis of an operation on both the first operation element and the second operation element, and mode C in which the acceleration, deceleration, and steering of the mobile object are controlled without being based on an operation on either the first operation element or the second operation element.

9. A mobile object control method comprising: recognizing, by a computer for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway, whether the mobile object is moving on the roadway or in the prescribed region; at least partially controlling, by the computer, a speed of the mobile object; and limiting, by the computer, the speed of the mobile object moving on the roadway to a first speed and limiting the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.

10. A storage medium storing a program for causing a computer for controlling a mobile object capable of moving both on a roadway and in a prescribed region different from the roadway to: recognize whether the mobile object is moving on the roadway or in the prescribed region; at least partially control a speed of the mobile object; and limit the speed of the mobile object moving on the roadway to a first speed and limit the speed of the mobile object moving in the prescribed region to a second speed that is lower than the first speed.
